SPHERE BRAWL: ROLLING RAMPAGE — TASK-09 ASSET BATCH 2: CHASSIS MODELS REPORT

Deliverable Package: Sphere_Brawl_Chassis_Batch2_Package.zip (23 Files + Root MANIFEST.json) | Max Vertex Radius <= 2.970 Studs | All 3 Chassis <= 250 Triangles | Cosmetic Parity (D-008)

Scope & Preservation Notice: This Batch 2 package delivers the three production-candidate player sphere chassis (*SM_Chassis_Trickster_01*, *SM_Chassis_Magnetar_01*, *SM_Chassis_Bulwark_01*) while preserving all existing Arena Tile and Stage 4 (`SM_Chassis_Rammer_01`, `SM_Hub_QueuePortalArch_01`) packages untouched. Every model uses the proven multi-node `.glb` (`Part_MAT_*`, `Part_EMI_*`) and multi-object `.obj`/`.mtl` architecture with baked `COLOR_0` vertex colors and an embedded 96×96 PNG palette texture atlas so colors are visible immediately on import into Roblox Studio.

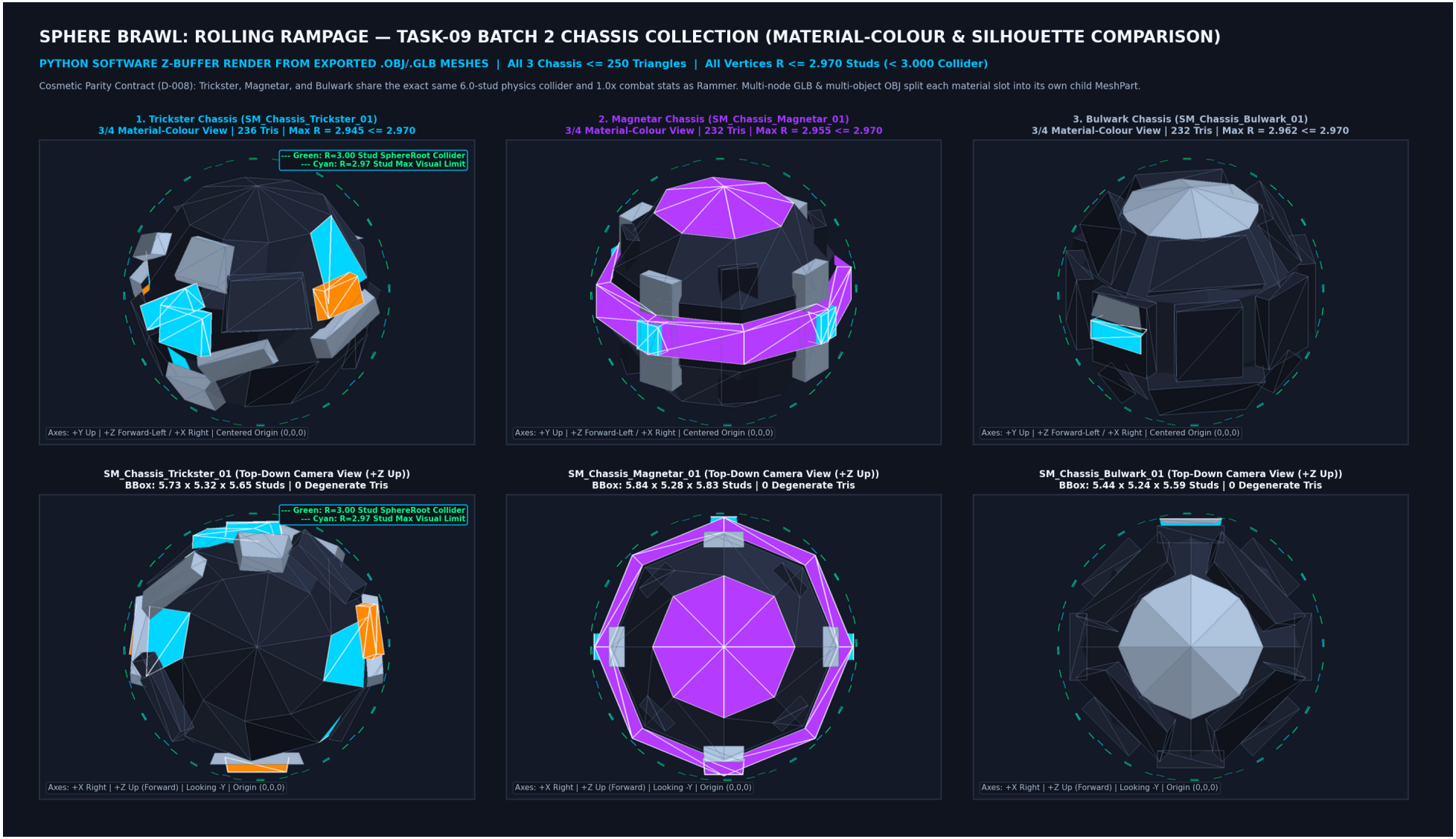
1. Batch 2 Chassis Models — Measured Dimensions, Radius Envelope & Triangle Counts

Chassis ID & Asset Name	Files (.obj / .mtl / .glb)	Tris (<=250)	Max Radius (<=2.970)	Bounding Box (W×H×D Studs)	Child MeshPart Nodes / Material Slots	File Check
Trickster SM_Chassis_Trickster_01	meshes/SM_Chassis_Trickster_01.obj meshes/SM_Chassis_Trickster_01.mtl meshes/SM_Chassis_Trickster_01.glb	**236** / 250	**2.9450** studs	5.73 × 5.32 × 5.65	Part_EMI_Cyan, Part_EMI_Kinetic, Part_MAT_AltNeutral, Part_MAT_DarkNeutral, Part_MAT_LightTrim	**PASS**
Magnetar SM_Chassis_Magnetar_01	meshes/SM_Chassis_Magnetar_01.obj meshes/SM_Chassis_Magnetar_01.mtl meshes/SM_Chassis_Magnetar_01.glb	**232** / 250	**2.9550** studs	5.84 × 5.28 × 5.83	Part_EMI_Cyan, Part_EMI_Magnetic, Part_MAT_AltNeutral, Part_MAT_DarkNeutral, Part_MAT_LightTrim	**PASS**
Bulwark SM_Chassis_Bulwark_01	meshes/SM_Chassis_Bulwark_01.obj meshes/SM_Chassis_Bulwark_01.mtl meshes/SM_Chassis_Bulwark_01.glb	**232** / 250	**2.9620** studs	5.44 × 5.24 × 5.59	Part_EMI_Cyan, Part_MAT_AltNeutral, Part_MAT_DarkNeutral, Part_MAT_LightTrim	**PASS**

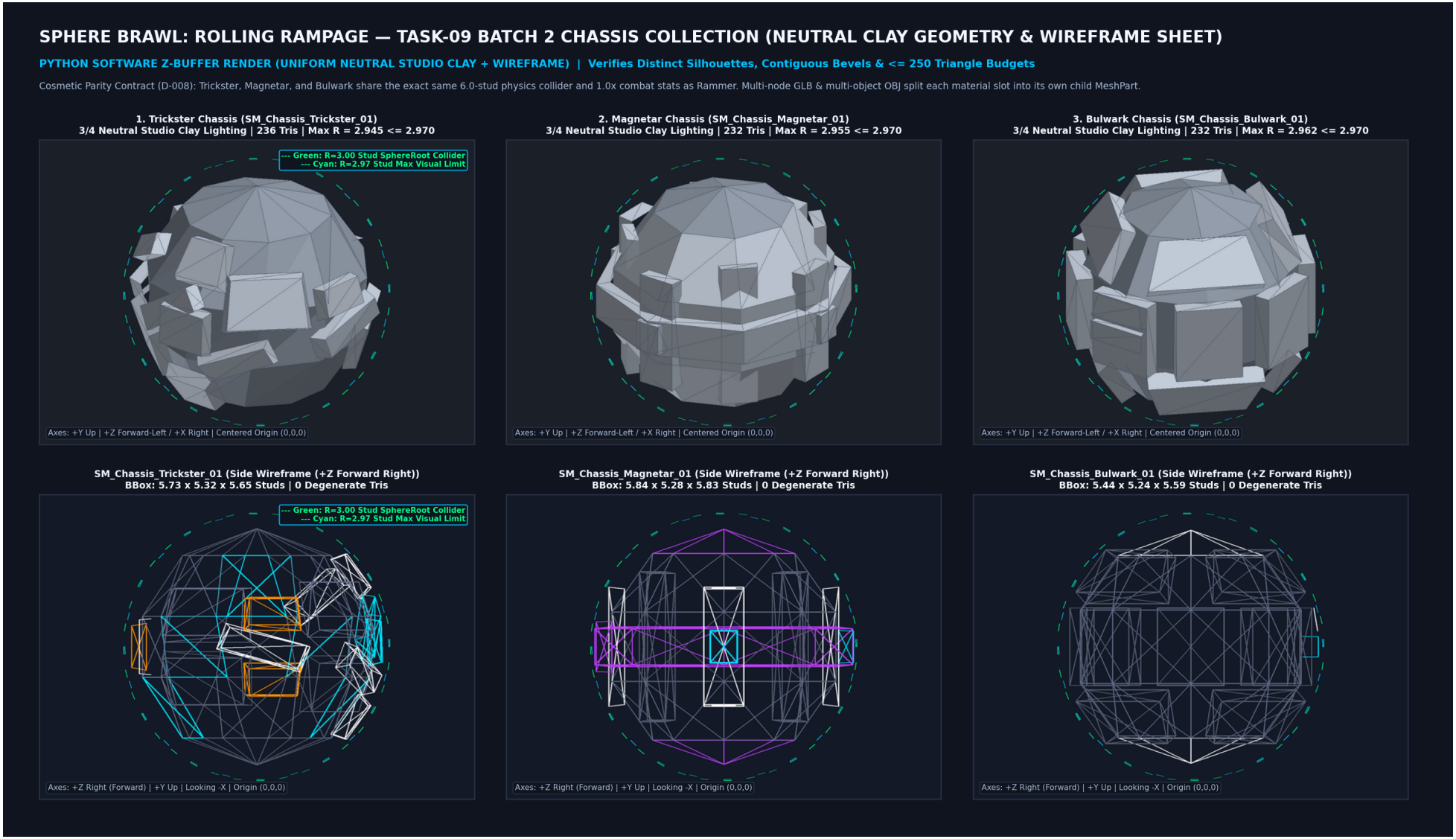
2. Validation Summary (Passed File Checks vs. Not-Run External Application Tests)

ID	Category	Check	Validation Method	Measured Result	Status
EXEC-01	1. Passed (Programmatic File Check)	**Sphere Collider Envelope (`Max Radius <= 2.970` Studs & Centered Origin)**	Computed Euclidean distance `sqrt(x^2 + y^2 + z^2)` across all vertices in `.obj` and `.glb` for Trickster, Magnetar, and Bulwark	Trickster max R = 2.94500 studs; Magnetar max R = 2.95500 studs; Bulwark max R = 2.96201 studs. 0 vertices exceed 2.97000 studs (0 exceed 3.00000-stud collider); all centered at (0,0,0) with +Y Up and +Z Forward.	**PASS**
EXEC-02	1. Passed (Programmatic File Check)	**Triangle Count Budget (`<= 250` Triangles Per Complete Chassis)**	Counted triangles in both `.obj` and `.glb` across all 3 complete chassis models	SM_Chassis_Trickster_01 = 236 tris; SM_Chassis_Magnetar_01 = 232 tris; SM_Chassis_Bulwark_01 = 232 tris. All 3 are strictly <= 250 triangles with 100% parity between `.obj` and `.glb`.	**PASS**
EXEC-03	1. Passed (Programmatic File Check)	**OBJ Syntax, Sub-Objects, Normals, Winding & Contiguous Geometry**	Line-by-line parser (`parse_and_validate_obj`) checking face indices, cross-product areas, and outward radial normals	0 syntax/index errors; 0 degenerate triangles (min area >= 0.0089 sq. studs); 0 normal/winding mismatches; all raised armor/fins are rooted at R = 2.60–2.64 studs inside the closed inner sphere hull (0 floating parts).	**PASS**
EXEC-04	1. Passed (Programmatic File Check)	**MTL Definitions, Approved RGB Palette & Atlas `map_Kd` Links**	Cross-checked every `.obj` `mtllib` and `usemtl` against its `.mtl` and the approved Sphere Brawl RGB palette (`parse_and_validate_mtl`)	3/3 `.mtl` files resolve; max `Kd`/`Ke` deviation from approved RGB is < 0.0001; every material links `map_Kd SphereBrawl_Chassis_PaletteAtlas.png`.	**PASS**
EXEC-05	1. Passed (Programmatic File Check)	**GLB Binary Validity, Multi-Node Child Parts, `COLOR_0` & Embedded PNG**	Parsed 12-byte GLB header + JSON + BIN chunks (`parse_and_validate_glb`) across all 3 `.glb` files	3/3 `.glb` files valid; each exports 1 root node + named `Part_*` child nodes per material slot, `COLOR_0` vertex colors, `TEXCOORD_0` UVs, and embedded 96x96 PNG palette atlas.	**PASS**
EXEC-06	1. Passed (Programmatic File Check)	**Deterministic Generator Reproducibility (Python & Blender Collector Parity)**	Re-ran `generate_chassis_batch2_pipeline.py` to `/tmp/det_chassis_test` and compared SHA-256 hashes; verified `DeterministicChassisCollector` in `generate_chassis_batch2_blender.py`	100% identical SHA-256 hashes across repeated runs; standalone Blender collector produces the exact same 236, 232, and 232 triangles.	**PASS**
EXEC-07	1. Passed (Programmatic File Check)	**Preview Image Readability & ZIP / Manifest Integrity**	PIL `Image.verify()` on all 5 PNG preview sheets + `zipfile.ZipFile.testzip()` CRC-32 check + SHA-256 verification against `MANIFEST.json`	5/5 PNG sheets verified; 0 corrupted ZIP entries; 100% match across all 23 files (12,412,539 B uncompressed, 10,672,610 B compressed).	**PASS**
EXT-01	2. Not Run (External Tool Unavailable)	**Headless Blender (`bpy`) Execution of `generate_chassis_batch2_blender.py`**	Checked Linux sandbox for `blender` binary and `bpy` module; verified standalone `DeterministicChassisCollector` in Python	Blender / `bpy` is not installed in this container. `generate_chassis_batch2_blender.py` is syntax- and collector-verified in Python, but executing `bpy` FBX/GLB export requires a workstation with Blender 3.x/4.x.	**NOT RUN (Requires Blender)**
EXT-02	2. Not Run (External Tool Unavailable)	**Roblox Studio 3D Importer Live Verification (`ASSET_PENDING` in Studio)**	Requires interactive Roblox Studio 3D Importer (`Import As Model = true`, `Merge Meshes = false`) in a separate test place	Not executed in this headless sandbox. Per TASK-09 & D-010, all three chassis remain `status = CANDIDATE_READY_FOR_STUDIO_IMPORT` (`robloxStudioVerified = false`) until imported and inspected by the owner in Roblox Studio.	**NOT RUN (Requires Roblox Studio)**

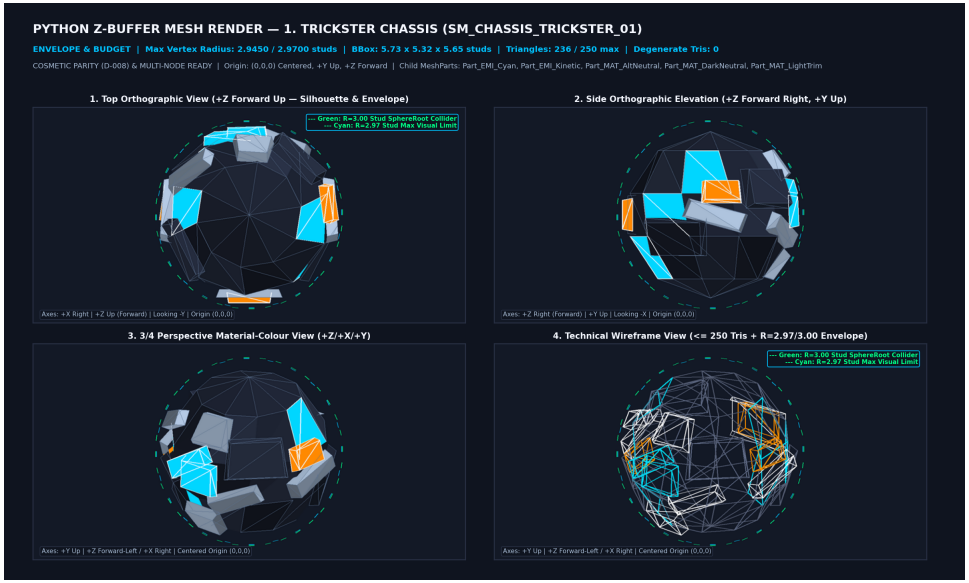
3. Batch 2 Chassis Collection — Material-Colour & Top-Down Silhouette Comparison

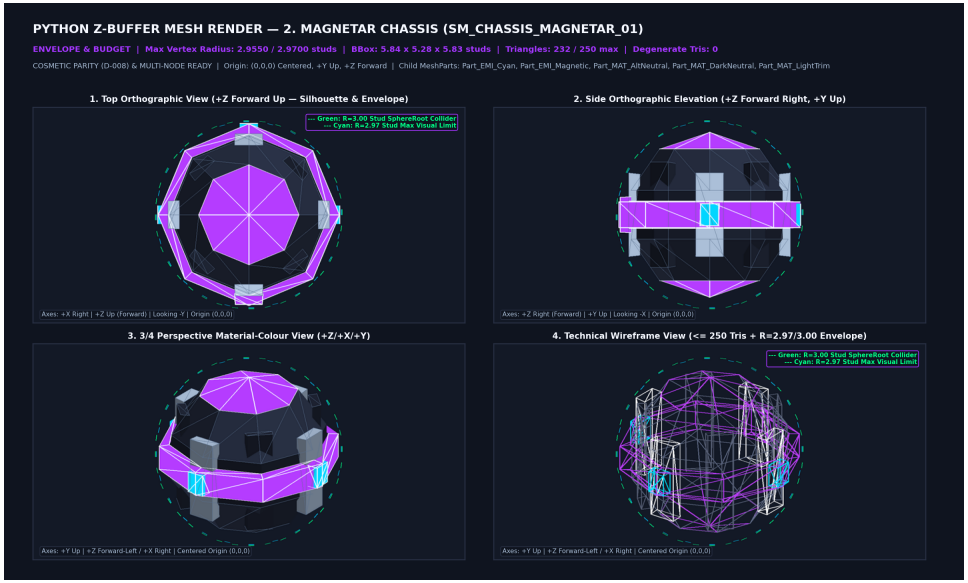


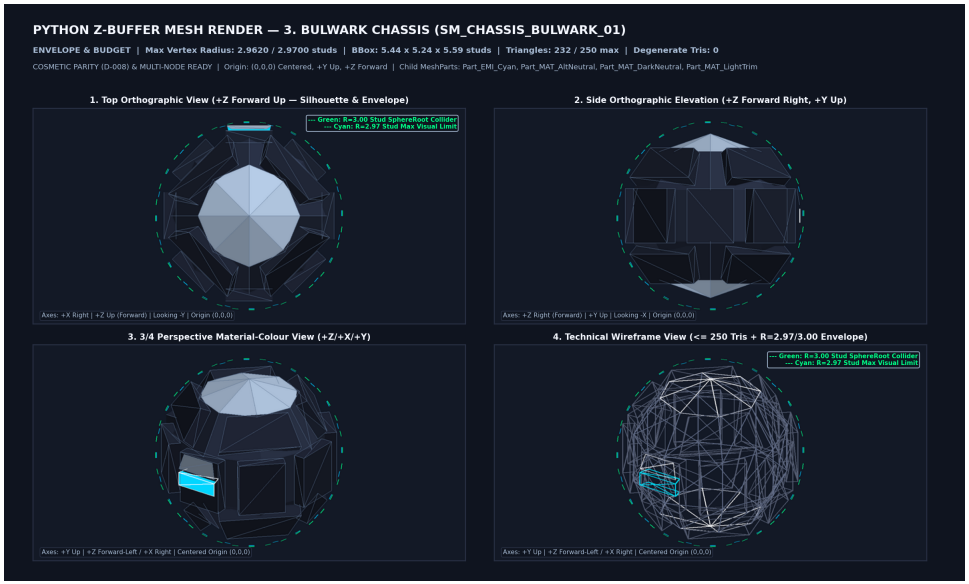
4. Batch 2 Chassis Collection — Neutral Studio Clay Geometry & Side Wireframe Comparison



5. Individual 4-Panel Technical Sheets — Trickster & Magnetar







Next Remaining Asset Batches for TASK-09:

• **Batch 3 — Starter Sphere Skins (4 Looks, D-020):** *Default* (unlocked), *Neon* (unlocked), *Gold* (locked previewable), and *Void* (locked previewable) material/surface definitions.

• **Batch 4 — Kinetic Hub Environment Station Models (§2.4, §11.3, TASK-03/09):** Customize Kiosk (4 pedestals), Shop/Cosmetics Stand, Mode Selection Board (*Free-for-All* + *COMING SOON*), Session Leaderboard Display, Queue Pad Ring/Floor Prompt, and Central Reactor Core rings (keeping Queue Portal Arch from Stage 4).

• **Batch 5 — Core Icons, Trails & Impact/Hazard VFX/Audio Placeholders (§11.5–11.6, D-021):** Cryo/Magnetic/Kinetic icons, *CoreStreak*/*CyanStreak* trails, *StandardSpark*/*CoreBurst* impact emitters, and verified/placeholder audio cues.